

Hybrid Image Compression Using DCT and Autoencoder

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Abstract—Medical image processing is an emerging area that has an effect on recognition, analysis and treatment techniques of the diseases. In medical image processing compression is required to reduce the bandwidth and storage related issues. In this method DCT and Autoencoder based image compression technique is used for the better compression of the medical brain images. To get the separate Region of Interest(ROI) and non-ROI based region Otsu method is considered . DCT and Autoencoder based hybrid compression is achieved on ROI and non-ROI regions of an image respectively. The evaluation parameter of this method are Peak Signal to Noise Ratio(PSNR),Structure Similarity Index matrix (SSIM),Feature Similarity Index Matrix (FSIM), Compression Ratio (CR) and Space saving(SS) percentage. For the medical brain images proposed method has PSNR of 44.45db , SSIM of 0.97 ,FSIM of 0.99, CR 9.26 and SS of 89.9(%).

Index Terms—Discrete Cosine Transformed(DCT), Autoencoder(AE), Otsu, Thresholding,Hybrid Image compression

I. INTRODUCTION

Medical image processing is a main part of digital image processing which deals with the medical treatment planning, medical science education, and research purpose [1]. For various medical treatment huge amount of image data is required and it is generated through the process of computed tomography (CT) and magnetic resonance images (MRI). There are some different way of medical imaging such as X ray, Ultrasound , Endoscopy etc. The data generated from the medical system is huge and it causes inappropriate bandwidth processing and storage requirements (senapati2016, Kumar et al. 2019) [2] [3]. To handle and process these huge amount of data in an efficient manner we need some compression techniques with less distortion. There are different techniques been proposed to handle these issues. In 2019 Ammah, et al. [4] proposed a picture archiving and communication system (PACS) medical imaging technique which provides low cost of storage with many resources. In 2020 UmaMaheswar et al. and rani et al. [5] proposed a compression techniques

for faster mode of transmission from a database and less memory based compression techniques with less transmission time. Different models of compression are there lossy and lossless compression [3]. For lossless mode of compression exact replica of the input image is received but it has less compression ratio. In case of lossy image compression the compression ratio is improved with some loss. To get the better compression with less loss two different techniques are combined known as hybrid compression technique. This is mainly used in medical image compression as it can give different level of compression for ROI and non ROI region image. The major challenges of medical image compression are it can't achieve higher rate of compression ratio with minimal data loss. To improve the medical image compression many different techniques are proposed. The DCTAE method has hybrid based image compression technique to improve the compression ratio with better visual quality.

Main contribution of the paper are

- An entropy based adaptive block selection process by using the Otsu based thresholding.
- Image adaptive DCT and auto encoder based hybrid image compression technique based on ROI and non-ROI regions is proposed.
- This method has improved the compression ratio by maintaining the quality of reconstruction while saving the space.

The residue of the paper are as follows section 2. Literature Review section 3. Proposed Method, Section 4. Simulation and Result analysis, Section 5. Conclusion.

II. LITERATURE REVIEW

There are many medical image compression techniques have been proposed. Some of the techniques are discussed below. Barros et al. 2016 proposed a lossless method of compression and fast transmission technique for the CT Perfusion.

This technique has a faster data transmission with the design complexity. sunil et al. [6] proposed compressive sensing medical image compression technique. Song et al. proposed a quantization technique to reduce the distortion caused by the high frequency pixel information. In 2018, Song et al. [7] proposed a Least Square based prediction technique for image compression by using the geometry adaptive partition. As the prediction of the block is based on each pixel it is a time consuming process. In 2018, Reddy et al. [8], proposed a compression technique which is a combination of three techniques those are discrete cosine transform, singular value decomposition and set partitioning in hierarchical trees. This work is not having a ROI and Non ROI mode of approach. In 2019, Ahilan et al. [9] proposed a ROI and non-ROI region based lossless image compression by using classification and blending prediction technique. Optimization and multi thresholding technique based block selection is done. In 2019 Manimekalai et al. [10] proposed a hybrid image compression technique for the medical image compression. In this method a hybrid mode of compression is performed by using two different techniques such as Lempel–Ziv–Welch and clipped histogram equalization based methods. In 2019, Devadoss and Sankaragomathi [11] proposed an effective technique to reduce the space saving of the medical image processing. In this method with the increase in the block size space saving is also increased. In 2020 Magar and Sridharan [12] proposed a hybrid mode image compression technique by using lossy and lossless mode of image compression. The border of the image is considered as non ROI region and the main portion of the image is considered as the ROI region. For compression non ROI region of image is compressed by using the lossy mode of compression and ROI region of the image compressed by using the lossless mode of compression. In 2021 Sabbavarapu et al. [13] proposed a DWT–RNN based image compression method for the medical images. In this method they have considered both the non-ROI and ROI image blocks compression with two different techniques such as RNN and DWT. To get the better space saving result the optimised RNN method by using PSO and GSA algorithms are used. This method has produced good result with the optimization but it is a complex process. In 2022, nayak et al. [14] proposed a compression technique by using the dual quantization based JPEG. Here the feature extraction is done by using the walsh hadamard kernel. There are different Hybrid image compression technique with better performance than the traditional codec. All methods represent the importance of space saving in order to reduce the storage space in the medical images [15].

III. PROPOSED METHOD

This section describes the proposed method and also how it deals with the recent issues of medical image compression. In medical image processing non-ROI and ROI region detection of the image play a major role for the compression. The quality of reconstruction for the ROI region of the image should be better in order to get proper diagnosis. This is an essential part as it contains more important information about

the patient's medical condition. So the rate of compression should be less for the ROI region. The background information of the image is known as non ROI region which can be more compressed to get the better bpp without affecting the quality of reconstruction.

In the proposed DCTAE method we have considered the block entropy to get the information content of the block. Again by using the Otsu [13] we have decided the threshold value to distinguish the non ROI and ROI region of an image. The ROI of the image is further process with the DCT based image compression to get the better information and an auto encoder based image compression technique for the non ROI region in image. This method is applicable to a number of images including the medical images. The DCTAE method work flow is illustrated in the Fig.1 as shown below. For the implementation purpose detailed block-wise description of the work flow is discussed in the below section.

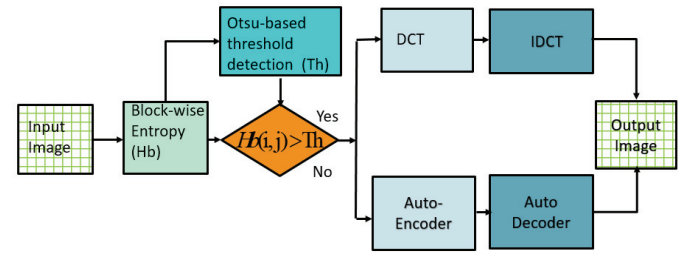


Fig. 1. Proposed Method Workflow

A. Entropy Detection

Spatial entropy of image is the predictable value of the information enclosed in some pixels. We compute the entropy H_{bi} of the i_{th} block as Eq.1.

$$H_{bi} = - \sum_{j=0}^{255} p_{ij} \log_2 p_{ij} \quad (1)$$

Where H_{bi} :entropy value of block and p_{ij} is the probability of ij pixel .Bit per pixel (bpp) is the unit of H_{bi} , and it represents the minimum number of bits to encode the block pixels without any loss.

The information content in a sparse components are related to the spatial entropy. Which indicates the spatial entropy of image block is inversely proportional to the sparse representation coefficients of block. In an image to maintain the quality of information with better bpp we need to provide a reduced amount of compression to the block with more information content and viceversa with less information content block. The normalized spatial entropy of each block is calculated by using Eq. 2.

$$w_i = \frac{H_{bi}}{\sum_{i=1}^L H_{bi}}, \quad (2)$$

The image block can be compress according to the entropy contrast. The thresholding of the spatial entropy values done by using the Otsu thresholding method. Based on the threshold value, block selection is done. Based on the threshold value the

selection of the blocks is done which segregate the blocks in two groups ROI and non-ROI blocks. After the block selection, the block needs to be encoded according to the information content. In the proposed method we have considered the DCT based image compression for the block with more information content and less information content image blocks are compressed with auto encoder. green DCT based method has more energy compaction than any other transform method, and it compacts all the low frequencies DCT coefficient by ignoring the high frequency components.As a result of which it gives more compression with quality of reconstruction.

B. Autoencoder based compression

An autoencoder is a neural network technique which is used for the compression. It has an encoder W , parametrized by θ , that computes a representation W' from the data x , and a decoder W' , parametrized by ϕ like sigmoid, tanh or the rectifier or linear units, that gives a reconstruction of image as X' of X . As Auto encoders can be used for de-noising or dimensionality reduction.In DCTAE we have considered the Autoencoder to compress the non informative region. When it is used for compression, the representation is done by using some quantization, leading to the new quantized representation or latent space representation. By training an auto encoder the reconstruction error can be minimized. By adding the sparsity constraint, we are in effect inducing regularization in our autoencoder, which can aid in training and computational efficiency. In DCTAE we have included the advantage of sparse autoencoder based minimization technique. Which has improved the space saving than the normal DCT based compression. The basic equation of the sparse auto encoder is shown in Eq.3.

$$x = W' \phi(Wx) \quad (3)$$

The basic auto encoder and the sparse autoencoder equation differ from each other by loss function as sparsity based loss function has one extra term known as the regularization term. The sparsity based loss function is given as below.

$$\min_{\theta, \phi} L(\theta, \phi) + \lambda L_{\text{sparsity}}(\theta, \phi) \quad (4)$$

where $\lambda > 0$ measures how much sparsity we want to enforce The implementation of proposed method is done by using the following Algorithm 1.

ALGORITHM:

- Step 1. Input an gray image of size $M \times M$ pixels
- Step 2. Divide the image into $k \times k$ non-overlapped blocks.
- Step 3. Finding the entropy of each block, $H_b(i, j)$
- Step 4. Get the threshold value by using the Otsu method from the entropy of the image blocks (Th)
- Step 5. Compare the image block entropy with the generated threshold value.
If $(H_b(i, j) > Th)$
Then flag 1\ Non Informative Block
else

flag 0\ Informative block
end if

- Step 6. The informative block is further processed with the DCT based image compression and non-informative block is processed with sparse-autoencoder to generate the encoded data.
- Step 7.Encoded data is decoded using IDCT and auto-decoder for the informative and non-informative blocks respectively to generate the reconstructed image.

IV. SIMULATION AND RESULT ANALYSIS

Quality assesment of the compression algorithms is done based on different evaluation parameters. Some important quality assesment parameters are considered here such as peak signal to noise ratio(PSNR),compression ratio, space saving (ss),Structurally simmilarity index(SSIM),feature similarity index (FSIM).PSNR is high for a good quality of reconstruction but it is not sufficient to evaluate the algorithm. So get the proper quality assesment we have considered all the structural and feature based similarity index whose value ranges between 0 to 1. The DCTAE algorithm is evaluated based on different datasets wit different format and size.The performance is measured based on 5 parameters PSNR (in dB), SSIM [16], FSIM [17],CR and SS. It can be seen that performance of the DCTAE is better than state of the art methods on the basis of 5 parameters. To evaluate the effectiveness of DCTAE we have considered 3 different sets of images those are standard, kodak and medical images [18]. The detail result analysis for the standard and Kodak dataset are evaluated and depicted in Table.I. From Table I it is seen

TABLE I
EVALUATION OF THE DCTAE METHOD BY USING STANDARD IMAGES AND KODAK IMAGES.

Images	PSNR	SSIM	FSIM	CR	Bpp
Lena	40.1787	0.9648	0.982	7.643	1.05
Airplane	39.0245	0.9558	0.983	8.203	0.98
Girl	36.5186	0.9234	0.966	7.403	1.08
Peppers	35.3759	0.9472	0.966	7.966	1.00
Boats	35.2241	0.9183	0.962	7.399	1.08
Girl-toy	34.4358	0.919	0.944	8.093	0.99
Kodak-Average	40.5678	0.9555	0.981	7.182	1.11

that the DCTAE has a higher PSNR,SSIM with better Bpp which indicates a better reconstructed image quality. For the practical application we have considered some of medical mri images as shown in the 3rd dataset and the detail result analysis is shown in Table.II.From Table.II it is illustrated that the DCTAE has an impressive performance for medical images.

TABLE II
EVALUATION OF THE DCTAE METHOD BY USING MEDICAL IMAGES.

	PSNR	SSIM	FSIM	CR
image1	44.83	0.99	0.99	9.50
image 2	39.12	0.95	0.98	8.89
image3	48.26	0.97	1.00	8.13
image4	45.32	0.96	1.00	7.90
image5	48.77	0.99	0.99	8.93
image6	47.56	0.98	0.99	10.00
image7	45.20	0.96	0.99	9.32
image8	45.46	1.00	1.00	10.36
image9	38.93	0.95	0.99	9.17
image10	43.47	0.95	0.98	9.27
image11	42.91	0.96	0.99	9.74
image11	46.05	0.97	0.99	9.48
image13	47.42	0.97	1.00	9.69
Average	44.87	0.97	0.99	9.26

The comparison of DCTAE method with different state of the art methods is given in the Table.III. The different state of the art methods considered for comparison are ADMG method based BTC [19], ImgAMBTC BTC based method [20] and WDCT is a DCT based method [14] which follows adaptive block wise compression. For all dataset images we have considered a 16×16 blocks size. For the impartial comparison all the state of the art methods are applied on the same image sets and performance is evaluated. PSNR, SSIM and FSIM values of the DCTAE are high in comparison with ImgAMBTC, ADMG with BTC and WDCT. It shows an enhancement in the visual quality of the reconstructed image. The detail of performance comparison is shown in the Table.III.

As these methods(img AMBTC,ADMG with BTC, WDCT) are not applicable for the medical images we have considered the standard images and Kodak data set images for the performance evaluation. The comparison of DCTAE method with the Img AMBTC, ADMG based BTC and WDCT method is shown in the Table.III. Although WDCT method has better performance, but it is not suitable for the medical images whereas DCTAE method can be applied to all types of images as illustrated from the Table.I, and II. SSIM comparison of different state of the art methods are shown in Fig.2.

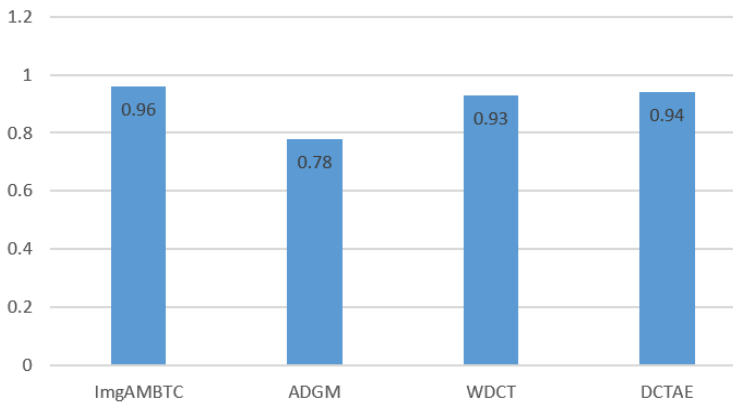


Fig. 2. Average SSIM value of different state of art method

Fig.3 illustrates the subjective evaluation of the DCTAE by using Lena, Airplane, Girl, Peppers, Girltoy, and Boats images. It is observed that, the DCTAE has better visual clarity than the other methods. For subjective evaluation of medical images, three samples of original and reconstructed brain images are shown in Fig.4.

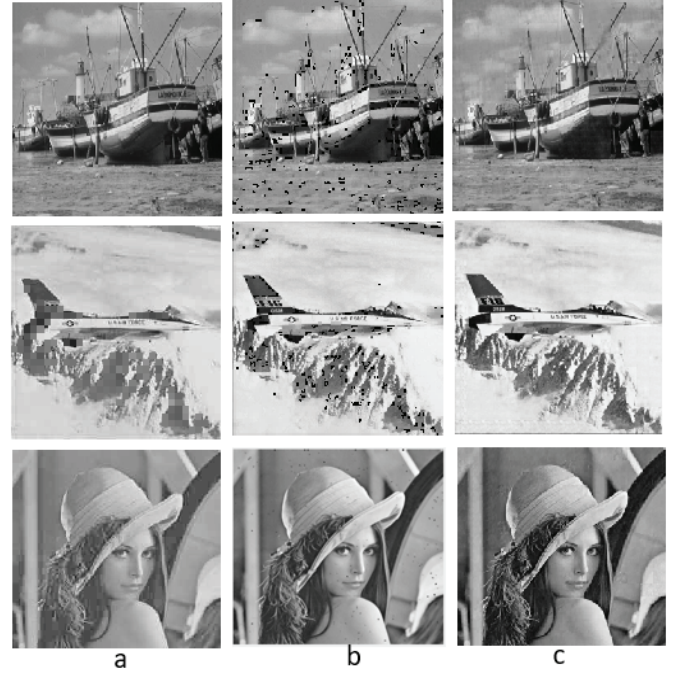


Fig. 3. Reconstructed Images by using a.wdct b.ADMG with BTC c. DCTAE

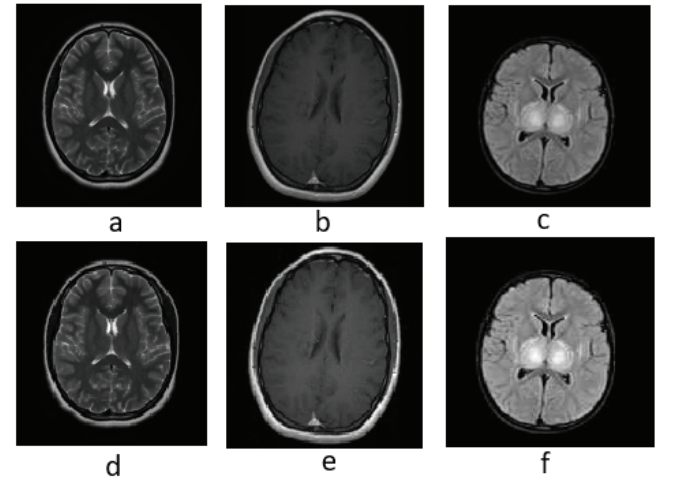


Fig. 4. (a, b, c) Original image. (d, e, f) Reconstructed brain image

The evaluation of DCTAE with state of the art methods in terms of PSNR, and CR and SS, are shown in the Table IV. From Table IV, it's concluded that the DCTAE accomplishes better PSNR, while keeping the proper CR as compared to the other methods. The FODPSO [9] has less PSNR, as it considers the entire image for compression regardless of non

TABLE III
COMPARISON OF THE DCTAE METHOD WITH STATE OF ART METHODS FOR STANDARD DATASET IMAGES.

Images	Img AMBTC [20]		ADMG [19]		WDCT [14]		DCTAE	
	PSNR	Bpp	PSNR	Bpp	PSNR	Bpp	PSNR	Bpp
Lena	38.89	2	35.61	1.8	40.8	1.23	40.2	1.1
Airplane	38.93	1.7	36.48	1.78	38.5	1.93	39.0	1.0
Girl	39.9	2.4	34.63	2.4	37.78	1.72	36.5	1.1
Peppers	39.1	2.4	35.6	1.93	36.54	1.24	35.4	1.0
Boats	37.98	2.4	34.97	2.05	38.8	2.03	35.2	1.1
Girtoy	39.01	2.4	33.65	1.79	37.6	2	34.4	1.0
Kodak Avg	39	2.2	32.82	2.2	42.9	1.6	40.6	1.1
Average	38.97	2.21	34.82	1.99	42	1.68	37.3	1.0

TABLE IV
PERFORMANCE COMPARISON OF THE DCTAE WITH STATE OF THE ART METHODS BY USING BRAIN IMAGES

	PSNR	CR	SS
FODPSO [9]	22.7223	2.13	–
QFOM [12]	33.51	24.61	–
BWT-MTF [11]	34.6905	4.634	78.42
DCTAE	44.87	9.26	89

ROI and ROI image. The PSNR of DCTAE is also higher than the QFOM [12] and BWT-MTF [11]. It shows that the DCTAE outperformed for the medical images than the other state of arts method. The DCTAE has a better PSNR and SS performances than the BWT-MTF. The SS of BWT-MTF method is 78.42 % which is less than the SS of DCTAE i.e., 89 %.

The use of the deep learning method for the image compression gives better performance than the traditional method. Precisely, the lossy compressions technique used in the image compression are having more compact representation with better quality of reconstruction. In the DCTAE method the combination of DCT and autoencoder gives better compression without hampering the quality of reconstruction.

V. CONCLUSION

In the proposed DCTAE based hybrid image compression algorithm the block entropy is considered to decide the blocklevel compression. The adaptive selection of the image block for the compression is based on the individual image. In this DCTAE we have implemented Hybrid concept of image compression by using DCT and Auto-encoder. Hybrid approach of Image compression using DCT and AE gives better PSNR with a good quality of reconstruction. To evaluate the performance of the DCTAE method many data sets with different image formats are used as well as a number of medical images are taken into consideration. DCTAE improves results competently by giving better SSIM [16] and FSIM

[17] of the reconstructed image. The DCTAE method has an average PSNR, SSIM, FSIM and CR value for the Standard dataset are 37.3, 0.94, 0.96 with a Space Saving value of 87 % and For the medical MRI images the PSNR, SSIM, FSIM and CR of the DCTAE method are 44.87, 0.97, 0.99, and 9.26 respectively. In the future we will implement the algorithm in a wide range of medical images. to get the better ROI and non ROI segmentation we will be considering some advance algorithms.

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